UNSUPERVISED REPRESENTATION LEARNING WITH DEEP CONVOLUTIONAL GENERATIVE ADVERSARIAL NETWORKS

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Abstract

The article "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks" introduces a class of convolutional neural networks known as deep convolutional generative adversarial networks (DCGAN). Through training on diverse image datasets, the authors demonstrate that both the generator and discriminator can learn hierarchical representations ranging from object parts to full scenes.

In this project, we attempted to build the DCGAN network architecture based on the model descriptions in the paper. To study whether the generator can learn a hierarchy of representations, we trained DCGAN on CelebA and evaluated generation quality through both visual inspection and controlled latent-space manipulations. The challenges include stabilizing GAN training and identifying representative latent vectors for semantic attributes such as smiling, wearing glasses, and face pose. Our results show that the original goal is partially accomplished: the "smiling" experiment successfully reproduces the patterns reported in the original work, and the "face pose" experiment demonstrates clear and consistent directional control. However, the "glasses" experiment exhibits weaker and less apparent effects, suggesting underfitting under the current training setup.

To investigate whether the discriminator learns hierarchical representations, we utilized the discriminator trained on ImageNet as a feature extractor for supervised classification tasks on the CIFAR-10 and SVHN datasets. We trained regularized linear L2-SVM classifiers on top of the learned convolutional features to evaluate their distinctiveness. The main technical challenges involved adapting the DCGAN architecture to handle 32x32 resolution inputs and implementing a pipeline to extract features from specific convolutional layers without disrupting the unsupervised training loop. The results demonstrate significant domain robustness; the model achieved a competitive classification accuracy on CIFAR-10 and a low error rate on SVHN (using only 1,000 labeled samples), confirming that the discriminator successfully learns semantically meaningful features applicable to downstream tasks even without supervised pre-training.

1. Introduction

Learning reusable visual representations from large-scale unlabeled image data is a fundamental challenge in computer vision, as supervised convolutional models typically require large amounts of labeled data. Motivated by the training instability of early generative adversarial networks and the limited effectiveness of previous unsupervised CNN-based approaches, DCGAN introduces architectural constraints that help stabilize adversarial training and enable the learning of meaningful hierarchical representations without supervision.

One goal of this project is to examine whether DCGAN trained on face images can learn semantically meaningful representations, as suggested in the original work. Specifically, we focus on reproducing the vector arithmetic experiments on facial samples and the face pose manipulation ("face turn") experiment. These experiments aim to evaluate whether changes in the latent representation correspond to semantic changes in the generated images, such as smiling or face orientation, without any explicit supervision.

A major technical challenge lies in reliably identifying representative latent vectors for specific semantic concepts. In practice, individual latent samples often exhibit high variability, making semantic manipulations unstable when using single examples. To address these challenges, we follow the original paper by averaging latent vectors over multiple exemplar images for each concept and performing vector arithmetic in the latent space. The effectiveness of the learned representations is then evaluated through visual inspection, focusing on consistency and interpretability of the semantic transformations.

Parallel to examining the generator, we also focus on the discriminator to determine if the network learns reusable feature hierarchies. Our objective is to utilize the discriminator not just for distinguishing real from fake images, but as a feature extractor for supervised classification on the CIFAR-10 and SVHN datasets. The main technical challenge involves adapting the DCGAN architecture to lower-resolution (32x32) inputs while maintaining feature richness and proving that the model can generalize to unseen datasets (SVHN) without any supervised fine-tuning. We solve this by implementing a pipeline that extracts activation maps from the discriminator's convolutional layers and training linear L2-SVM classifiers on these features. This approach allows us to quantitatively assess the semantic quality of the unsupervised representations and verify the domain robustness claims made in the original paper.

2. Summary of the Original Paper

2.1 Methodology of the Original Paper

The classic generative adversarial networks (GAN) framework consists of a generator and a discriminator trained adversarially using convolutional neural networks. Taking a random noise vector as the input, the generator is to generate a new image, while the discriminator attempts to distinguish between real images and generated samples. Rather than introducing new loss functions, the paper proposes several architectural constraints that improve convergence and representation learning of the classic GAN. These constraints are applied consistently across datasets and experiments.

After constructing the DCGAN architecture, the authors train the model on multiple datasets for different experimental purposes.

First, the authors train the DEGAN on a face dataset scraped from web image searches of people's names (Model 1). Model 1 is mainly used for latent space vector arithmetic experiments, which investigate how well the generator captures semantic features in the latent space, such as facial concepts.

Second, a DCGAN model trained on the ImageNet-1K dataset (Model 2) is used as a feature extractor for image classification tasks. Through this setup, the authors demonstrate that the discriminator learns useful and transferable feature representations. Specifically, the authors propose an evaluation protocol where the trained discriminator weights are frozen and used as a fixed feature extractor. Feature maps are extracted from the convolutional layers and max-pooled to produce a 4 × 4 spatial grid, which is then flattened to form a feature vector. To quantify the quality of these representations, a regularized linear L2-SVM classifier is trained on top of them. The paper validates this methodology on two downstream tasks: classification on the CIFAR-10 dataset to measure overall accuracy, and on the Street View House Numbers (SVHN) dataset to assess domain robustness and performance under limited labeled data constraints

2.2 Key Results of the Original Paper

2.2.1 Key Results of Face Experiments

A key result reported in the paper is that semantic manipulations in the latent space leads to meaningful and consistent changes in generated face images. By computing arithmetic on the Z vectors of sets of exemplar samples — such as smiling, wearing glasses, or face pose—the authors demonstrate that adding or subtracting these vectors produces predictable visual transformations.

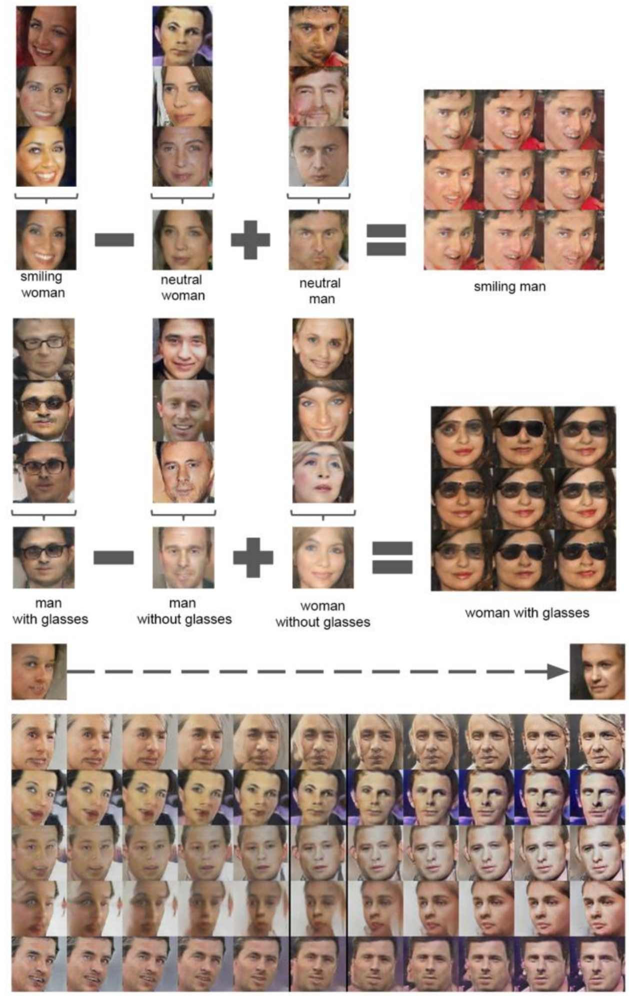

Figure 1. results of model 1 in original paper

2.2.2 Key Results of Imagenet Experiments

For the model 2 trained on ImageNet, the authors evaluate the quality of the learned representations by using the discriminator as a feature extractor for supervised classification.

Key result on CIFAR-10 as shown in table1: The paper reports that a linear L2-SVM classifier trained on top of the discriminator's convolutional features achieves a classification accuracy of 82.8%. This performance demonstrates that the unsupervised DCGAN features are highly discriminative and outperform several other unsupervised learning baselines.

Table 1. CIFAR-10 classification results using pre-trained model. DCGAN is not trained on CIFAR-10, but on Imagenet-1k, and the features are used to classify CIFAR-10 images.

Model	Accuracy
1 Layer K-means	80.6%
3 Layer K-means Learned RF	82.0%
View Invariant K-means	81.9%
Exemplar CNN	84.3%
DCGAN + L2-SVM	82.8%

Table 2. SVHN classification with 1000 labels.

Model	Error rate
KNN	77.93%
TSVN	66.55%
M1+KNN	65.63%
M1+M2	36.02%
M1+TSVM	52.33%
DCGAN (ours) + L2-SVM	22.48%
Supervised CNN	28.87%

Key result on SVHN as shown in table 2: To test domain robustness, the authors applied the same feature extraction method to the Street View House Numbers (SVHN) dataset. Using only 1,000 labeled training examples, the DCGAN + L2-SVM pipeline achieved a test error rate of 22.48%. This result is particularly significant as it rivals the performance of fully supervised CNNs trained on the same limited data (which achieved a 28.87% validation error), confirming that the discriminator learns robust hierarchical representations that generalize well even with minimal supervision.

3. Student Project Methodology

3.1. Objectives and Technical Challenges

3.1.1.Face Experiments

The objective of face experiments is to verify the vector arithmetic phenomenon through three face-related experiments. Specifically, we conduct experiments on facial expression (smiling experiment), presence of glasses (glass experiment), and face pose rotation (face pose experiment) to determine whether performing vector arithmetic on the z vectors corresponding to images of specific semantics can change the generated image's semantics in a particular direction. In simple terms, the smiling experiment attempts to obtain a smiling male face by applying vector arithmetic in the form of “smiling woman − neutral woman + neutral man”. Similarly, the glasses experiment follows the same arithmetic pattern to add or remove eyeglasses, while the face pose experiment uses vector arithmetic to rotate the face orientation.

These experiments face several challenges. First, acquiring semantic vectors requires manual selection of representative samples, such as identifying smiling versus neutral faces or faces with and without glasses, which may introduce subjectivity and sampling bias. Second, facial attributes are often not completely disentangled in real-world data, which means that modifying one semantic concept (e.g. smiling) may unintentionally affect other attributes (e.g. hairstyle, skin tone). In addition, the effectiveness of vector arithmetic is sensitive to training quality, and under-fitting of the generator can lead to weak or inconsistent semantic transformations.

3.1.2 Imagenet Experiments

The main objective of imagenet experiments is to evaluate the discriminator's feature learning quality using quantitative metrics, rather than just visual inspection. We use the trained discriminator as a fixed feature extractor to perform classification tasks on CIFAR-10 and SVHN datasets. The goal is to verify if the features learned from unlabeled ImageNet data are useful enough to train a simple linear SVM classifier with high accuracy.

We faced three main technical challenges in this part. First is the resolution mismatch: CIFAR-10 and SVHN images are 32×32, which is smaller than the standard 64×64 input. We had to adjust the discriminator’s structure (depth and stride) to prevent feature map issues. Second is the limited data constraint: for the SVHN experiment, we are restricted to using only 1,000 labeled samples. We had to implement a careful data sampling method to ensure a fair evaluation and avoid unstable results. Finally, there is the domain gap: we need to prove that features learned from natural objects (ImageNet) can successfully generalize to digit images (SVHN) without any fine-tuning.

3.2.Formulation and Design Description

3.2.1 Problem formulation and design description of face experiments

Let $z \in \mathbb{R}^d$ denote a latent noise vector sampled from a prior distribution, $G(z)$ denote the trained DCGAN generator. The generator G maps noise vector z to an image x:

$$x = G(z) \tag{1}$$

For a given semantic concept (e.g., smiling), we define a semantic direction in the latent space as:

$$v_{direction} = \frac{1}{N_1}\sum_{i=1}^{N_1} z_i^{(1)} - \frac{1}{N_2}\sum_{j=1}^{N_2} z_j^{(2)} \tag{2}$$

where $z^{(1)}$ and $z^{(2)}$ correspond to the vectors associated with images that possess or do not possess the target semantic concept, respectively.

A modified latent vector is then computed via vector arithmetic:

$$z_{modified} = \bar{z} + \alpha v_{direction} \tag{3}$$

And then we can generate the target image:

$$x_{target} = G(z_{modified}) \tag{4}$$

3.2.1.1 Design of smiling experiment

The smiling experiment aims to transfer the “smiling” concept across gender (from woman to man) while preserving facial identity. To be more specific:

$$v_{smile} = \frac{1}{3}\sum_{i=1}^{3} z_i^{(smiling\ woman)} - \frac{1}{3}\sum_{j=1}^{3} z_j^{(neutral\ woman)}$$

$$z_{smiling\ man} = \overline{z_{neutral\ man}} + \alpha v_{smile}$$

Here, $\overline{z_{neutral\ man}} = \frac{1}{3}\sum_{i=1}^{3} z_i^{(neutral\ man)}$, $\alpha = 1$.

Then, the target image is generated as:

$$x_{smiling\ man} = G(z_{smiling\ man})$$

In simple terms, this experiment follows the vector arithmetic:

$$\overline{z_{smiling\ woman}} - \overline{z_{neutral\ woman}} + \overline{z_{neutral\ man}} = z_{smiling\ man}$$

3.2.1.2 Design of glass experiment

The glasses experiment evaluates whether the presence of eyeglasses can be added or removed through vector arithmetic. This experiment follows the same vector arithmetic formulation with a different semantic direction:

$$\overline{z_{man\ with\ glasses}} - \overline{z_{man\ without\ glasses}} + \overline{z_{woman\ without\ glasses}} = z_{woman\ with\ glasses}$$

3.2.1.3 Design of face pose experiment

The face pose experiment investigates whether face orientation corresponds to a controllable direction in the latent space. It slightly differs from first two experiments in that the coefficient α changes from -2 to 2 to visualize continuous pose transitions along the semantic direction.

$$z_{face\ orientation} = \overline{z} + \alpha v_{facing\ right}$$

In summary, the three face-related experiments are implemented under a unified vector arithmetic framework using the same trained DCGAN generator. By treating semantic manipulation as simple linear operations in the latent space and applying the same system design across all experiments, we are able to clearly evaluate how well DCGAN learns semantic representations.

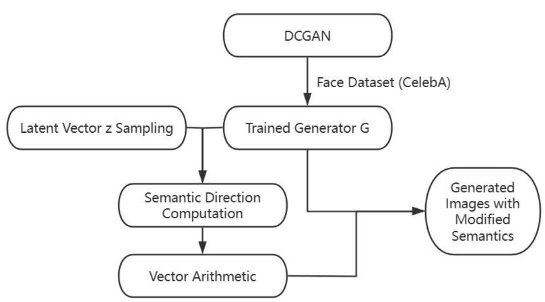

Figure 2. block diagram of face experiments

3.2.2 Design of Feature Extraction for Classification

The design for imagenet experiments focuses on repurposing the discriminator from a binary classifier (Real vs. Fake) into a feature extractor for multi-class classification. Unlike the vector arithmetic experiments in face experiments, this design involves a two-stage pipeline described in section 3.2.1.1 and 3.2.1.2.

3.2.1.1 Design of Feature Extraction

After unsupervised training on the ImageNet dataset, we set the weights of the discriminator D. For any input image x (from CIFAR-10 or SVHN), we extract the feature maps from the last convolutional layer before the output. In our adapted 32 × 32 architecture, this corresponds to a spatial grid of activations. Let $\phi(x)$ represent this feature extraction function:

$$\phi(x) \in \mathbb{R}^{4\times4\times256} \quad (5)$$

This 3D tensor is then flattened into a 1D vector v of dimension 4096:

$$v = \text{flatten}\big(\phi(x)\big) \quad (6)$$

3.2.1.2 Design of Linear Classification

We then train a Linear Support Vector Machine (L2-SVM) on top of these feature vectors. The SVM attempts to learn a weight matrix W and bias b to predict the class label y:

$$y_{pred} = \arg\max(W^T v + b) \quad (7)$$

In simple terms, this experiment treats the discriminator as a "frozen eye" that converts images into numeric codes. We then test if a simple linear calculator (SVM) can look at these codes and correctly identify whether the image is a plane, a bird, or a house number. For the SVHN experiment specifically, we strictly limit the training data to 1,000 balanced samples to rigorously test the robustness of these generated codes.

4. Implementation

4.1 Data

4.1.1 CelebA Dataset

In the original DCGAN paper, the authors use a face dataset scraped from random web image searches of people's names, which is not released publicly. Therefore, we use the CelebA dataset as a substitute for training the DCGAN model used in the face experiments. CelebA serves as a widely accepted substitute for the original face dataset for several reasons. First, both datasets consist of large-scale collections of human face images collected from the web, exhibiting significant diversity in identity, expression, pose, and accessories. In addition, CelebA has been extensively used in prior work on generative models, including GAN-based face generation and latent space analysis.

CelebA contains over 200,000 celebrity face images with rich variations in facial attributes and provides aligned and cropped faces, making it well suited for generative modeling. In our experiments, all images are resized to 64×64 and normalized to the range [-1,1], following the same data preprocessing setup used in the original DCGAN paper.

4.1.2 Mini-Imagenet Dataset

We used the MiniImageNet dataset with images size 64 × 64 for the unsupervised training of the DCGAN discriminator. This dataset serves as a large-scale source of natural images, enabling the model to learn rich, hierarchical visual features without using any label information. To match our modified DCGAN architecture, all training images were resized to 32×32 resolution and normalized to the range [-1,1] before being fed into the discriminator.

4.1.3 CIFAR-10 Dataset

The CIFAR-10 dataset was used to evaluate the classification accuracy of the learned features. It consists of 60,000 32×32 color images across 10 classes (e.g., airplanes, birds, cats), split into 50,000 training and 10,000 test images. Since CIFAR-10 shares the same resolution as our resized ImageNet data but contains different object distributions, it serves as an ideal benchmark for testing general feature quality.

4.1.3 SVHN (Street View House Numbers) Dataset

To evaluate the domain robustness of our model, we used the SVHN dataset, which contains 32×32 cropped images of house numbers obtained from Google Street View. Unlike ImageNet and CIFAR-10 which contain natural objects, SVHN focuses on digits (0-9). Consistent with the original paper's protocol, we strictly limited our usage to 1,000 labeled training examples (100 per class) to test the model's performance under semi-supervised conditions with limited data.

4.2 Deep Learning Network

The architecture of the DCGAN model is adjusted based on the classic GAN model. According to the description in the text, the DCGAN framework we built is shown in the figure 3 and figure 4.

The objective function of DCGAN follows the standard formulation of generative adversarial networks. Specifically, the discriminator D and generator G are optimized according to the following objective:

$$\min_G \max_D \mathbb{E}_{x\sim p_{data}}[log(D(x))] + \mathbb{E}_{z\sim p(z)}[log(1 - D(G(z)))]$$

Here, x denotes a real image sampled from the data distribution p_{data}, and z denotes a latent noise vector sampled from a prior distribution $p(z)$, which is chosen to be a uniform distribution in our project.

In order to train both the train the model, the objective function is implemented using separate loss functions for the discriminator and the generator. The generator is trained to fool the discriminator by making generated images as close to realistic distribution as possible. The generator loss is defined as:

$$G_{loss} = -\mathbb{E}_{z\sim p(z)}[log(D(G(z)))]$$

The discriminator is trained to distinguish real images as real and generated images as fake. Its loss function is defined as:

$$D_{loss} = -\mathbb{E}_{x\sim p_{data}}[log(D(x))] - \mathbb{E}_{z\sim p(z)}[log(1 - D(G(z)))$$

Both the generator and the discriminator are optimized using the Adam optimizer, which is commonly adopted in GAN training due to its stability and adaptive learning rate properties.

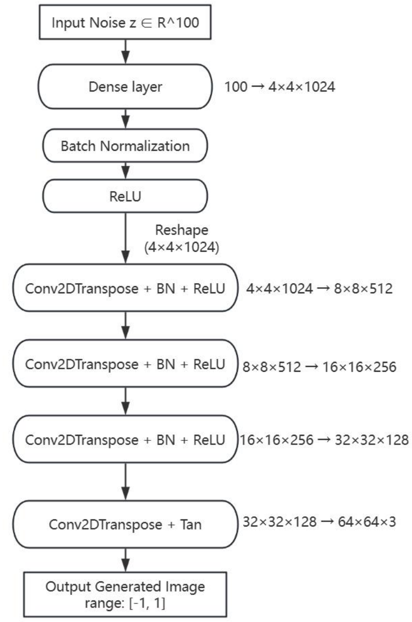

Figure 3. generator of DCGAN

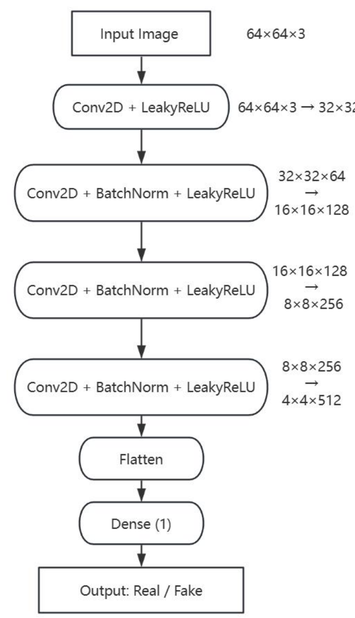

Figure 4. discriminator of DCGAN

The DCGAN model is trained using an alternating optimization strategy, where the discriminator and generator are updated sequentially within each training iteration. The training process is implemented based on binary cross-entropy loss with logits and optimized using the Adam optimizer.

At each training step, a mini-batch of latent noise vectors $z \in \mathbb{R}^{100}$ is sampled from a uniform distribution U(-1,1). The generator takes the noise

vectors as input and produces a batch of fake images. The discriminator then evaluates both real images from the dataset and generated images from the generator.
The discriminator loss is computed using both real and fake predictions, encouraging it to assign higher scores to real images and lower scores to generated images. The generator loss is computed based on the discriminator's predictions of fake images, encouraging the generator to produce images that the discriminator classifies as real.

Table 3. hyperparameters of DCGAN model

hyperparameters	value
learning rate	2×10^{-4}
Adam momentum parameter β_1	0.5
batch size	128

Gradients of the discriminator loss and generator loss are computed separately using two independent gradient tapes. The discriminator parameters are updated first, followed by the generator parameters. This alternating update strategy allows both networks to improve simultaneously through adversarial learning. During training, loss values for both the generator and discriminator are recorded at each epoch. Sample images generated by the generator are periodically saved to monitor training progress. Model weights are also saved at the end of each epoch for further analysis.

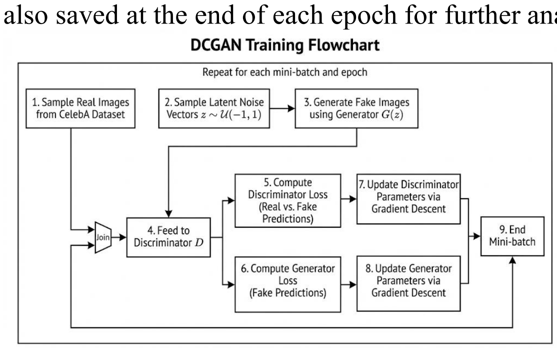

Figure 5. DCGAN Training Flowchart

4.3 Software Design

4.3.1 Top level flow chart

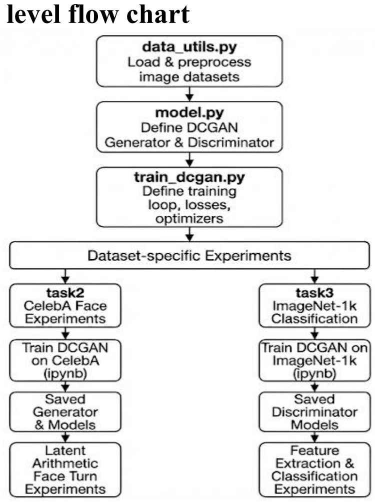

Figure 6. Top level flowchart

4.3.2 Detailed Component

Table 4. Detailed component of our project

File	Description
model.py	Defines the core DCGAN architecture, including the generator and discriminator networks. Both models are implemented as convolutional neural networks following the architectural guidelines proposed in the original DCGAN paper, such as strided convolutions, batch normalization, and appropriate activation functions. These model definitions are shared across all experiments and datasets.
data_utils.py	Contains utility functions for loading and preprocessing image datasets. This includes reading raw image files, resizing images to the target resolution, normalizing pixel values to match the generator's output range, and constructing TensorFlow datasets for efficient batching and shuffling.
train_dcgan.py	Implements the training logic of DCGAN, including the loss functions for the generator and discriminator, optimizer definitions, and the training loop for adversarial optimization. This file encapsulates the core training procedure and is reused by different experiments to ensure consistency across datasets.
task2: face experiments (CelebA)	The task2 directory corresponds to face generation and latent-space analysis experiments using the CelebA dataset. In celeba_training.ipynb, the DCGAN model is trained on face images, and the trained generator and discriminator weights are saved. Subsequent experiments are conducted within the same directory, including: latent vector arithmetic (e.g., smiling vs. neutral), and face pose manipulation using the learned "face turn" vector. The quality and interpretability of the learned latent representations are evaluated through visual inspection of generated images.
task3: ImageNet Experiments	The task3 directory corresponds to the unsupervised representation learning experiments using the ImageNet datase. It stores the final trained model weights: generator_task3_imagenet.h5, discriminator_task3_imagenet.h5. The key experimental workflow is implemented in imagenet_feature_extraction.ipynb. In this notebook, we train the generator32 and discriminator32 and the pre-trained discriminator is then loaded and set fixed to serve as a feature extractor. The notebook performs the quantitative evaluation by: Extracting convolutional features from the CIFAR-10 and SVHN datasets. Training Linear SVM classifiers on these features. Calculating classification accuracy and error rates to verify the quality of the learned representations (reproducing Table 1 and Table 2 of the original paper).

Overall, the project follows a modular design in which model architecture, data processing, and training logic are decoupled, while dataset-specific experiments are conducted in separate task directories to facilitate reproducibility and comparative analysis.

5. Results

5.1 Project Results

5.1.1 Results of Face Experiments

5.1.1.1 Generation Quality

We use the model trained on the CelebA dataset to randomly generate 16 human face images and visually assess the quality of the generated images.
In figure 6, we can see that the model produces face images with coherent global structure and reasonable visual diversity, demonstrating successful learning of high-level facial representations. While local artifacts and blurriness remain, the generation quality is sufficient for analyzing semantic properties of the latent space.

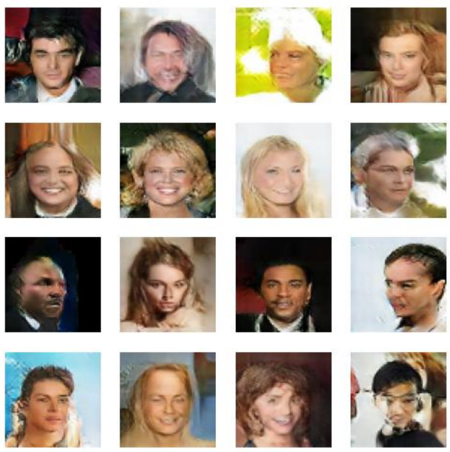

Figure 6. Images generated from DCGAN trained on CelebA

5.1.1.2 Vector Arithmetic

To reproduce the vector arithmetic experiment on facial attributes, we first randomly sampled 500 latent noise vectors z and generated the corresponding face images using the trained generator. These images were then manually inspected, and samples that clearly matched the desired semantic categories were selected.

5.1.1.2.1 Smiling Experiment

Three representative images were chosen for each category: smiling woman, neutral woman, and neutral man(figure 7). Following the vector arithmetic procedure introduced in section 3.2.1, a new latent vector was computed and fed into the generator to produce images corresponding to a smiling man. To examine robustness, small random perturbations were added around this resulting vector, and multiple samples were generated(figure 8).

Figure 7. Manually selected face images used to construct averaged latent vectors for the smiling woman, neutral woman, and neutral man categories.

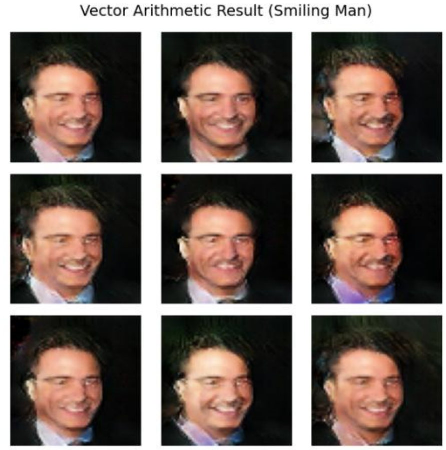

Figure 8. Generated face images representing a smiling man, obtained by applying latent vector arithmetic.

The resulting generated faces clearly exhibit smiling expressions while preserving male facial characteristics, indicating that the "smiling" attribute is encoded in the latent space in a way that is approximately linear across genders. This observation is consistent with the findings reported in the original DCGAN paper.

5.1.1.2.2 Glass Experiment

The vector arithmetic experiment for the glasses attribute follows the same procedure as the smiling experiment.
Compared to the smiling experiment, the results of the glasses experiment are apparently weaker. While some generated images exhibit faint visual patterns resembling glasses, the attribute is often blurred, or mixed with other facial features such as lighting, eye regions. The semantic transformation does not appear as stable as the smiling attribute.

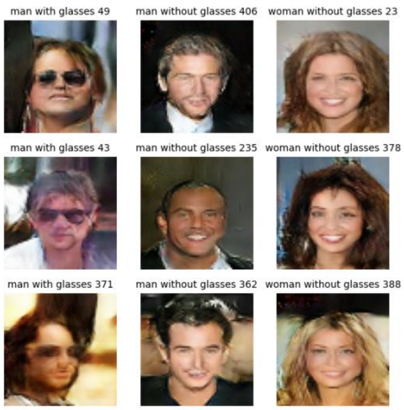

Figure 9. Manually selected face images: man with glasses, man without glasses, and woman without glasses categories.

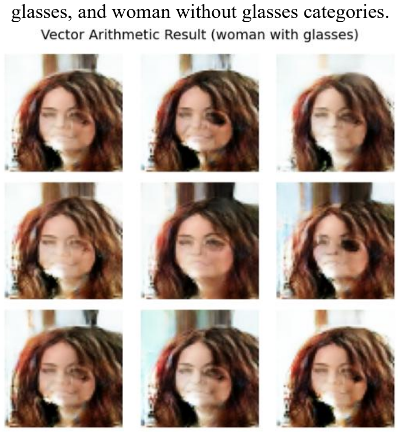

Figure 10. Generated face images representing a woman with glass.

To further investigate this behavior, we increase the number of exemplar images used to compute the averaged latent vectors, selecting ten samples per category instead of three (see the ten samples in appendix). The resulting images are shown in Figure 11. Although this adjustment slightly improves stability, the overall effect remains limited: glasses are still inconsistent, and the generated images frequently exhibit indistinct facial details around the eye region. This suggests that simply increasing the number of samples is insufficient to improve generation quality.

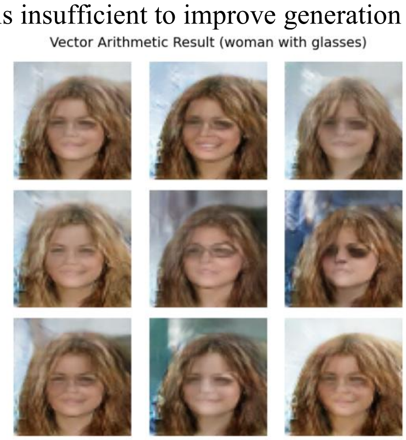

Figure 11. Generated face images representing a woman with glass (10 samples).

5.1.1.2.3 Face pose Experiment

Manually select four face images looking left and four images looking right, as shown in the figure 12. The corresponding latent vectors are averaged within each group to obtain representative “left-looking” and “right-looking” face vectors (figure 13), and the difference between them is used as a face pose (turn) vector.

Using this pose vector, we apply latent-space manipulation to randomly generated samples. As shown in the figure 14, adding the pose vector results in faces that gradually change their head orientation from left to right, while largely preserving identity and other facial attributes. These results indicate that face pose can be effectively manipulated through simple vector arithmetic.

Figure 12.(left) Manually selected face images (looking left vs looking right). Figure 13.(right) Average face of looking left vs looking right.

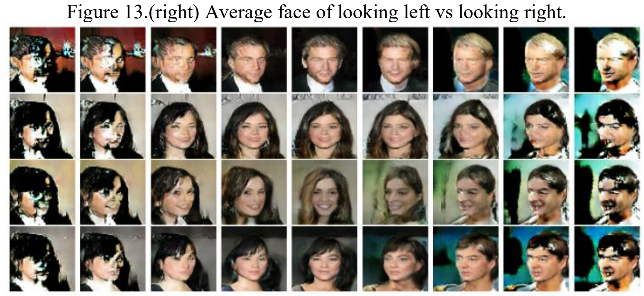

Figure 14. Face pose manipulation using latent vector arithmetic.

5.1.2 Results of Imagenet Experiments

In this section, we quantitatively evaluate the quality of the representations learned by the discriminator. Unlike the visual inspection used for the generator, we assess the experiment by repurposing the frozen discriminator as a feature extractor for downstream classification tasks.

5.1.2.1 CIFAR-10 Classification Performance

To measure how well the discriminator captures semantic features from natural images, we trained a regularized Linear L2-SVM on the features extracted from the CIFAR-10 dataset.

As shown in Figure 15, our model achieved a classification accuracy of 57%. This performance significantly exceeds the random guess baseline (10%), confirming that the discriminator has learned a hierarchy of features that are semantically meaningful and distinct, rather than just memorizing low-level pixel statistics.

```
==================================================
FINAL RESULTS - TASK 3 (Section 5.1 Reproduction)
==================================================
Model: Discriminator32 (As Feature Extractor)
Dataset: CIFAR-10
Accuracy: 57.00%
==================================================
Paper Baseline: 82.8% (Note: This baseline used the full ImageNet dataset
   and more training epochs).
==================================================
```

Figure 15. Final Results of task3 on CIFAR-10

5.1.2.2 Domain Robustness on SVHN

To verify the "domain robustness" of the learned features, we applied the same feature extraction pipeline to the SVHN dataset, which differs significantly from the ImageNet training domain. Following the original paper's protocol, we restricted the classifier training to only 1,000 labeled samples.

As presented in Figure 16, our model achieved a test error rate of 40.8%. This low error rate demonstrates the model's strong generalization ability, proving that unsupervised features learned from natural objects can be effectively transferred to digit recognition tasks even with minimal supervision.

```
==================================================
FINAL RESULTS - TASK 3 Part 2 (SVHN Table 2)
==================================================
Model:      DCGAN Discriminator + L2-SVM
Dataset:    SVHN (1000 Labeled Samples)
Accuracy:   59.92%
Error Rate: 40.08%
==================================================
Paper Result: 22.48% Error Rate.
==================================================
```

Figure 16. Final Results of task3 on SVHN dataset

5.2 Comparison of the Results Between the Original Paper and Students' Project

In the original DCGAN paper, the experiments were conducted using a large-scale face dataset and trained for a substantial number of iterations on high-performance GPUs. In contrast, our project trains DCGAN on the CelebA dataset and the Mini-ImageNet dataset for representation learning. Unlike the original paper which utilized the full ImageNet-1k dataset (over 1.2 million images), we employed Mini-ImageNet (60,000 images across 100 classes, ~7GB) due to storage constraints on our GCP instance. With a reduced number of training epochs due to limited computational resources and time constraints. As a result, the overall performance of our model is below the original′s, and training time in our experiments is significantly shorter than that reported in the original paper(1 hour on CelebA, and approximately 2 hours on Mini-ImageNet).

Overall, the face experiments lead to phenomena that are largely consistent with those reported in the original paper, although there remains a noticeable gap in generation quality and stability. The smiling experiment achieves results comparable to the original work, while the glasses experiment fails to clearly reproduce the same semantic patterns. For the face pose experiment, the head rotation process is generally clear, but the image quality degrades to some extent during the transformation.

Regarding the imagenet experiments, we observe a performance gap between our reproduction and the reported benchmarks. For CIFAR-10 classification, our model achieved an accuracy of 57%, whereas the original paper reported 82.8%. Similarly, for the SVHN domain robustness task, our method yielded an error rate of 40.08%, compared to the paper's 22.48%.

5.3 Discussion and Insights Gained

For face experiments, one possible reason for the performance gap is the difference in datasets, as the original paper uses a larger face dataset with more training samples. In addition, limited computational resources restrict the number of training epochs. We follow the original hyperparameter settings (e.g. batch size of 128), but are only able to train the model for 10-15 epochs, which may result in underfitting. Moreover, the imbalance of certain attributes in the CelebA dataset, such as the relatively small number of women wearing glasses, may further affect the quality of the learned representations.

For classification task, the quantitative gap between our results and the original paper can be attributed to three primary factors:

- Data scale and diversity: the original paper trained on the full ImageNet-1k dataset (1.28 million images), whereas we were constrained by GCP storage limits to use Mini-ImageNet (60,000 images). Deep learning models, particularly GAN, require vast amounts of data to learn robust, generalized features. A 20-fold reduction in training data inevitably leads to less diverse filter representations, reducing the feature extractor's discriminative power on downstream tasks like CIFAR-10.
- Training duration and stability: due to computational time constraints, our model was trained for fewer

epochs compared to the extensive training schedule in the original work. GAN is difficult to converge; with limited training steps, the discriminator may not have fully matured to capture high level semantic abstractions, resulting in lower classification accuracy.

- Resolution constraints: We trained on 32x32 resolution images to match the Mini-ImageNet and CIFAR-10 dimensions. The original DCGAN architecture was optimized for 64x64 inputs. The reduction in spatial resolution means the convolutional layers have less information to process, potentially discarding fine-grained textures that are useful for classification.

Despite these limitations, the fact that our model achieved an error rate of 40% on SVHN with only 1,000 labels is significant. It confirms that even with a smaller dataset (Mini-ImageNet), the DCGAN unsupervised learning objective is powerful enough to learn features that generalize across domains (from natural objects to digits).

6. Future Work

Future work could focus on improving both the stability and quality of face generation by training the model for more epochs and leveraging more powerful computational resources. Exploring larger or more balanced face datasets may further help the model capture fine-grained attributes such as glasses more effectively.

In addition, to bridge the gap between our reproduction results and state-of-the-art generative models, future iterations could scale the experiments in both data size and model capacity. One immediate extension would be training on the full ImageNet-1k dataset, which would require access to high-performance computing clusters with sufficient storage and multi-GPU support to handle the large-scale data and increased batch sizes. Furthermore, extending the model to higher image resolutions, such as 64×64 or 128×128, could enable the network to learn richer spatial hierarchies. This would involve adding additional upsampling layers to the generator and corresponding convolutional layers to the discriminator.

7. Conclusion

Overall, the face experiments largely reproduce the qualitative findings of the original DCGAN paper, particularly for smiling and face pose manipulation, while highlighting limitations in modeling localized attributes such as glasses. These results suggest that DCGAN can learn meaningful facial representations, but its performance is sensitive to training scale, data balance, and attribute complexity. The imagenet experiments showed that the discriminator, trained without any supervision, learns hierarchical features that are transferable to downstream classification tasks. Although our quantitative results on CIFAR-10 and SVHN were limited by the use of the smaller Mini-ImageNet dataset, the model still demonstrated strong domain robustness.

8. Acknowledgement

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7. References

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8. Appendix

8.1 Ten samples per category for glass experiment

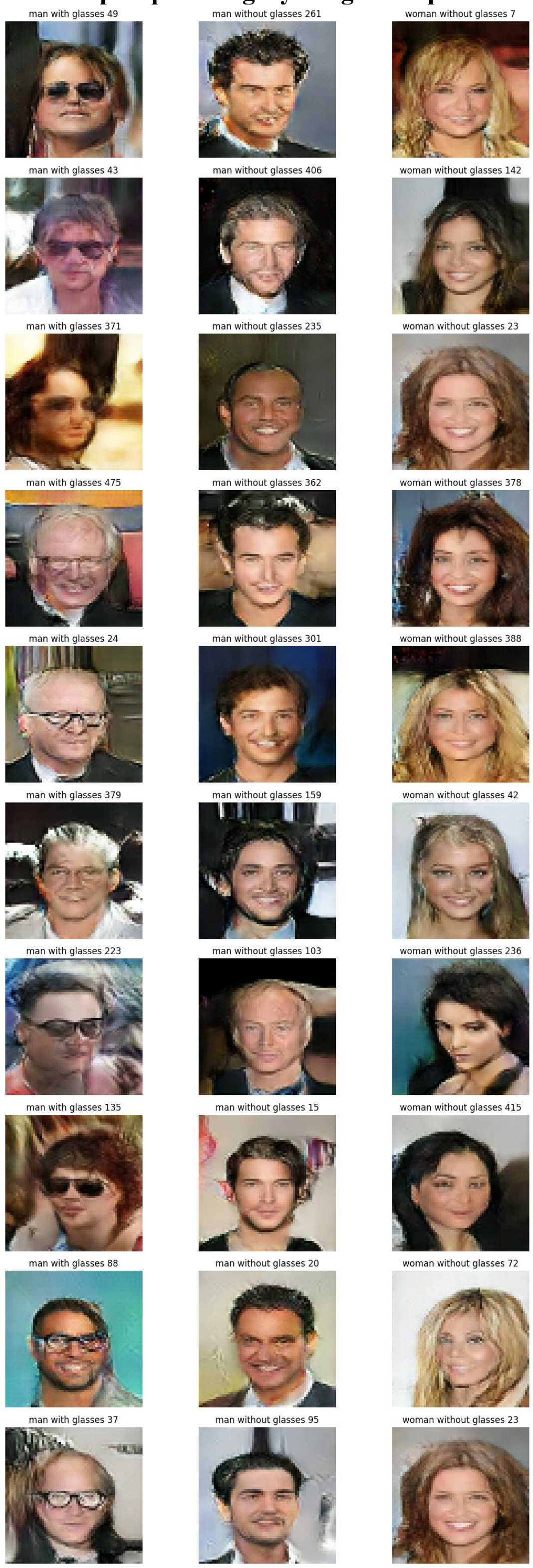

Figure 17. 10 selected face images used to construct averaged latent vectors for the man with glasses, man without glasses, and woman without glasses categories.

8.2 Individual Student Contributions in Fractions

	UNI1	UNI2	UNI3
Last Name	Bolun Li bl3147	Ziyi Zhao zz3459	Natel Putinati Diego dn2659
Fraction of (useful) total contribution	40%	40%	20%
What I did 1	Constructed the DCGAN generator and discriminator models. (model.py file on Github)	Constructed the DCGAN generator32 and discriminator32 models. (Especially for size 32 x32 dataset in model.py file on Github) and write the data loader function including (CIFAR-10 SVHN MiniImagenet in data_utils.py)	Write some part of data_utils.py code
What I did 2	Successfully reproduce experiments on CelebA dataset (Github task 2)	Successfully reproduce experiments on dataset miniImagenet, CIFAR-10 and SVHN (Github task3)	Try to reproduce experiments on dataset LSUN (github task 1)
What I did 3	Write and edited the project report	Write and edited the project report	No contribution to the writing part